

AI Workout Planner

Feature Breakdown

This feature set has been chosen as the core foundation based on a thorough revision of successful competing products in the fitness app market. The selected features focus on personalization, AI-driven workout generation, progress tracking, and user engagement, ensuring a competitive and feasible scope for the 7-week project timeline.

1. Core Features Overview

Feature	Description	Implementation Method	Feasibility (7 Weeks)
User Profile	Stores user data (age, weight, goals, equipment)	SQLite	Very feasible
AI Workout Generator	Generates personalized workouts	API call to OpenAI/Gemini	Very feasible
Exercise Library	List of exercises with instructions	SQLite	Very feasible
Progress Tracking	Logs completed workouts	SQLite	Feasible
AI Chat Assistant	Fitness Q&A chatbot	API call to AI model	Feasible
Calorie Tracker (Optional)	Manual calorie entry or AI-estimated calories	Manual input or AI prompt	Feasible (manual) / Stretch (AI)

2. User Profile

Aspect	Details
Purpose	Personalize workouts and AI responses by tailoring workout plans and recommendations based on user-specific data. Enables targeted fitness guidance and improved user engagement.
Data Fields	Name, age, height, weight, goals, experience level, available equipment, and preferences. Critical for accurate workout customization.
Data Inputs	User entered profile data during onboarding and profile updates. Optionally sync with external health data sources for updates.
Data Outputs	Customized workout plans and AI responses personalized to user profiles. Data used to adapt workout difficulty and recommendations.
Storage	SQLite for persistent local storage. Data encryption recommended for privacy.
Implementation	Use SQLite for data storage with a user login for verification. Ensure data validation, secure handling, and efficient data retrieval.
UI Considerations	Simple, intuitive profile editing screens with clear input fields, validation feedback, and progress indicators. Responsive design for various devices.
Optional Enhancements	Integration with external health data sources (Google Fit, Apple Health) for automatic updates. Add profile analytics dashboard as stretch goal.
Notes	Keep UI simple and editable. Prioritize data accuracy, privacy, and user trust. Anticipate future scalability for additional data fields.

3. AI Workout Generator

Aspect	Details
Purpose	Generate personalized workout plans tailored to user goals, fitness level, and preferences using AI to optimize effectiveness and engagement.
Input	User profile data (age, weight, goals, equipment) and selected workout goals or focus areas.
Output	AI-generated structured workout plan including exercises, sets, reps, rest periods, and progression suggestions.
Method	API call to OpenAI/Gemini with prompt templates designed to elicit detailed, personalized workout plans. Possible use of TensorFlow pre-trained models for cost and availability reasons (Unsure of difficulty)
Implementation	Integrate AI API calls within app backend. handle prompt construction, response parsing, and error handling.
UI Considerations	Display generated workouts in an easy-to-follow format with clear instructions, timers, and progress tracking. Allow user customization and feedback.
Optional Enhancements	Add adaptive workout adjustments based on user feedback and progress. incorporate multimedia exercise demos.
Notes	No ML training required relies on external AI API. Fast to implement but dependent on API availability and cost.

4. Exercise Library

Aspect	Details
Content	20–40 exercises with instructions
Data Source	Local JSON file
UI	list + detail screen
Optional	Link to YouTube demos

5. Progress Tracking

Aspect	Details
Purpose	Track and log user workout completion, duration, and notes to monitor fitness progress and motivate continued engagement.
Logged Data	Workout name, date, duration, notes, calories burned, heart rate (optional), and user feedback.
Storage	SQLite database
Data Inputs	User inputs workout completion details and optionally syncs wearable data (heart rate, calories).
Data Outputs	Progress history, summaries, and visual charts for user review.
UI	Simple history list with filters and sorting, optional charts and progress

	graphs as stretch goals.
Implementation	Use local database APIs for data storage and retrieval, implement UI components for history and charts, handle data syncing from wearables if available.
Optional Enhancements	Add detailed analytics, goal tracking, and integration with external health platforms (Google Fit, Apple Health).
Notes	Feasible within 7 weeks. syncing wearable data may require additional permissions and API integration effort.

6. AI Chat Assistant

Aspect	Details
Purpose	Provide real-time fitness advice, answer user questions, and offer motivational support through conversational AI to enhance user engagement and personalized guidance.
Data Inputs	User queries, context from user profile, workout history, and previous chat interactions to tailor responses accurately.
Data Outputs	Contextual, relevant fitness advice, explanations, and motivational messages delivered via chat interface.
Method	API call to AI model (OpenAI/Gemini) with prompt engineering to maintain conversational context and accuracy.
Implementation	Integrate AI API calls within chat backend, manage session state, prompt

	construction, response parsing, and error handling.
UI Considerations	Simple, intuitive chat interface with support for text input, quick replies, and conversational history. Responsive design for mobile and desktop.
Optional Enhancements	integration with workout data for personalized suggestions
Notes	Shares AI infrastructure with workout generator, requires careful prompt design to avoid generic or off-topic responses.

7. Calorie Tracker (Optional)

Aspect	Details
Purpose	Enable users to monitor and manage their daily calorie intake to support fitness goals such as weight loss, maintenance, or muscle gain.
Data Inputs	Manual calorie entries by users for meals and snacks, optionally meal descriptions for AI estimation.
Data Outputs	Daily calorie totals, meal-wise calorie breakdowns, and trends over time to inform user decisions.
Implementation	Manual entry via simple input forms, AI estimation via API calls to AI models with meal description prompts.
UI Considerations	Clear input fields for manual entry, intuitive display of calorie summaries and trends, and feedback on calorie goals.
Optional Enhancements	Integration with barcode scanners for quick entry AI meal photo recognition

Notes	Manual entry is straightforward and feasible. AI estimation is a stretch goal requiring robust AI prompt design and validation.
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8. Project Feasibility Summary

Feature	Difficulty	Time Required	Recommended
User Profile	Easy	3–4 days	Yes
AI Workout Generator	Easy	1 week	Yes
Exercise Library	Easy	1 week	Yes
Progress Tracking	Medium	1 week	Yes
AI Chat Assistant	Medium	1 week	Yes
Calorie Tracker	Easy–Medium	1 week	Optional

9. Recommended 7-Week Timeline

Week	Focus Area
Week 1	Planning, UI wireframes, AI API testing

Week 2	User Profile + basic navigation
Week 3	AI Workout Generator
Week 4	Exercise Library
Week 5	Progress Tracking
Week 6	Calorie Tracker + AI Chat Assistant
Week 7	Polish, testing, presentation prep